

# Delphi App Translation Studio

## Engineering Guide

How the product is built, why it is built that way,  
and what holds it together

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Frameworks: Delphi VCL and FireMonkey · Platform: Windows (Studio)

Written from the source. Every number, file name and line reference in this document was taken from the code as it stands on the date above.

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# 1. Purpose and audience

This guide is for the engineer who has to change this product: to fix a defect, add a capability, or judge whether a proposed change is safe. It assumes fluency in Delphi and no prior knowledge of this codebase.

It is not a user guide. It does not explain how to run the Setup Wizard; it explains what the Wizard is doing and why.

Where a design decision was reached by measurement rather than by reasoning, the measurement is given. Where something is known to be wrong, it is said so plainly and pointed at the Fix List rather than glossed over. A guide that only describes the parts that work is not an engineering document.

## 2. What the product does

Delphi App Translation Studio takes a Delphi application and produces the files needed to run that application in another language — including the layout adjustments the new language requires, because translated text is rarely the same size as the original.

Concretely, it:

- reads the application's forms and source and finds the text a user will see;
- works out what each string *means* in that application, without a person typing anything;
- translates through DeepL or Google using the developer's own API key;
- measures the translated text with the same engine the framework will draw it with, and plans the width, height, wrapping, font size and position changes it needs;
- writes an offline JSON language pack;
- and applies that pack at run time through a component, so the user can change language while the application is running.

The deployed application needs no API key and no internet connection.

## 3. The five principles

Almost every design decision in this codebase follows from one of five commitments. When a change seems to conflict with the surrounding code, it is usually because it conflicts with one of these.

### 3.1 The target application is evidence, never a workpiece

The translator reads the target project's `.dfm`, `.fmx`, `.pas`, `.dpr` and `.dproj` files and never writes to them. Every artefact it produces lives outside the target tree. This is not a configuration option; it is the reason a

developer can trust the tool with a working application.

It has a consequence that recurs throughout this guide: where a translation genuinely requires a change to the application — a `TranslateText` call for a string the application composes itself — the tool's job is to *say so precisely*, not to make the change.

### 3.2 The runtime is offline

Translation happens once, at development time. What ships is a JSON pack read from disk. No key, no network, no telemetry. This separates the one phase that needs the internet from the one that must never depend on it.

### 3.3 Artefacts are readable and diffable

The catalog, the layout proposal and the runtime pack are JSON. They can be read in a text editor, compared between versions, committed to source control and corrected by hand. Several defects described in this guide were diagnosed by reading those files rather than by debugging, which is an argument for the format on its own.

### 3.4 One decision lives in one place

Learned the hard way. The list of layout properties a pack may carry once existed in **four** units. They drifted, and each drift silently deleted a feature: the planner decided correctly, the runtime would have applied correctly, and the pack in between simply did not carry the value. There was no error anywhere.

That list now lives only in `DAT.Runtime.LanguagePack`, which the exporter and both applicators already share. Section 14 explains the tests that now guard the joins.

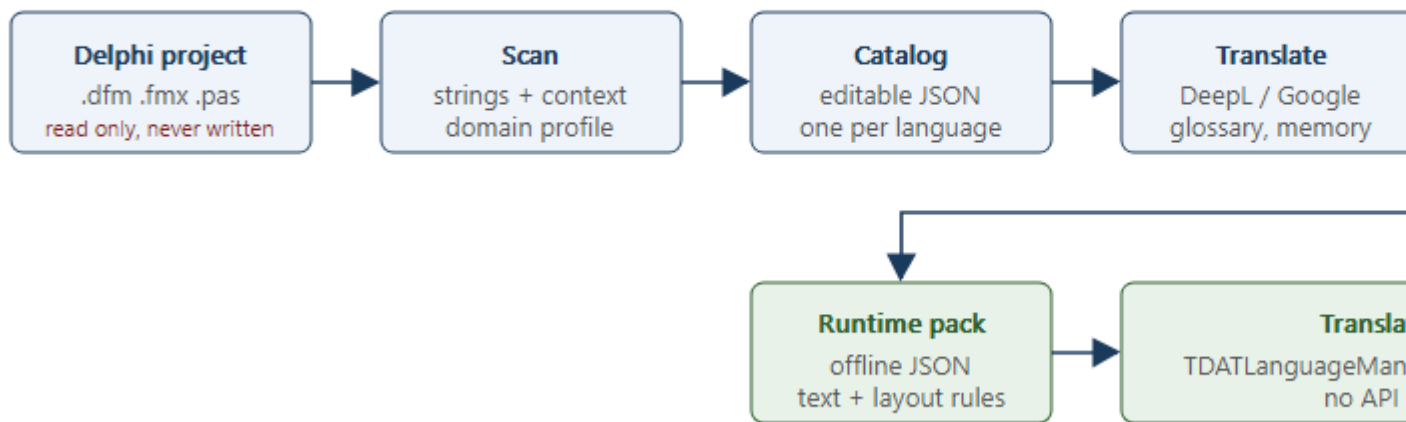
### 3.5 Both frameworks, one implementation

VCL and FireMonkey differ in defaults, in property names and in what the framework does for you. The response is not two code paths but one implementation behind a seam. The clearest example is text measurement (section 10.1); the clearest payoff is right-to-left mirroring (section 11), where the VCL has native support, FireMonkey has none at all, and both are served by the same pass.

## 4. Architecture

**Figure 1 — The translation pipeline**

Each stage writes a durable artefact. Nothing downstream reads the target application.



Each stage writes a durable artefact and the next stage reads it. That is what makes the pipeline diagnosable: when something is wrong on screen, the question is always *which artefact first contains the mistake*, and every one of them can be opened and read.

### 4.1 Source layout

| Folder           | Units | Lines | Responsibility                                                               |
|------------------|-------|-------|------------------------------------------------------------------------------|
| source\core      | 10    | 3,497 | Catalog types, JSON, workspace paths, glossary, hyphenation, pack builder    |
| source\scan      | 10    | 3,741 | Reading forms and Pascal; context and domain profiling                       |
| source\review    | 5     | 4,720 | The layout planner, text measurement seam, code-owned geometry               |
| source\provider  | 8     | 1,371 | DeepL and Google: placeholders, batching, language codes, retry, credentials |
| source\runtime   | 5     | 4,452 | What ships inside the customer's application                                 |
| source\component | 7     | 2,064 | TDATLanguageManager and the language selectors                               |

|                    |   |       |                                      |
|--------------------|---|-------|--------------------------------------|
| s                  |   |       |                                      |
| source\studio      | 4 | 5,204 | The Studio and the Setup Wizard      |
| source\integration | 5 | 2,172 | Component kits, packages, deployment |
| source\validation  | 1 | 403   | The catalog validator                |

The division that matters most is **runtime versus everything else**. Units under source\runtime and source\components are compiled into the customer's application. They must be small, must not reach for the network, and must never write to a user's disk uninvited. Everything else runs only in the Studio.

## 5. The scan

DAT.Scan.Project drives the scan; DAT.Scan.FormText reads designer files and DAT.Scan.PascalResources reads Pascal.

### 5.1 What is collected, and what is deliberately not

The scan is limited to project-referenced source units and designer resources. Broad harvesting of Items.Add, Lines.Add, TextOut and similar calls is **disabled on purpose**: in practice those carry data rows, file names, log lines and generated content rather than stable interface text. An earlier version harvested them and produced thousands of non-UI strings from one application.

Designer-authored Items and Lines are still covered, because those come from the form file where they belong.

### 5.2 Collections

Grid columns are the awkward case and are worth understanding before changing the parser. A TDBGrid keeps its heading on the column's Title, so the catalog key is grdSchedule.Columns[0].Title.Caption — one level further in than the column itself. Collections nest, so a collection's end must not pop the object stack; getting that wrong corrupts every control after the grid.

### 5.3 Text ownership

Every string is classified by who controls it at run time. The classification decides what happens later and is worth stating exactly:

| Classification    | Meaning                                                          | Applied automatically?                      |
|-------------------|------------------------------------------------------------------|---------------------------------------------|
| designerAutomatic | Set in the form file and not touched by code                     | Yes                                         |
| runtimeWired      | Assigned in code, but through a path the runtime can reach       | Yes                                         |
| runtimeUnwired    | Assigned in code, composed at run time                           | <b>No</b> — needs a call in the application |
| applicationData   | A data value, not interface text                                 | No                                          |
| suspicious        | Looks like data, or like something that should not be translated | No                                          |

The third row is the one that surprises people. An application that builds a status bar in code — `StatusBar1.Panels[1].Text := 'Items in list: ' + ItemCount.ToString;` — overwrites anything the pack writes there moments later. The classification is correct and the translation is genuinely impossible without a `TranslateText` call in the application. What the product does *not* yet do is tell the developer which strings those are and where they live, though it holds the file and line for each. That is on the Fix List.

## 6. Context: telling the translator what a string means

A translation service sees one string at a time. Given *Play Date From* and told only that it is "text used in a desktop application", DeepL returned *Spielverabredungen* — an afternoon arranged between children — for a column of scheduled start times. Given *Close*, Spanish returned *cerca*, meaning *nearby*.

Everything needed to settle both was already in the scan. Nobody had to type it.

### 6.1 The application domain profile

DAT.Scan.DomainProfile reads the application rather than recognising it. An earlier version matched six subjects from keyword lists — music, scheduling, business records, clinical records, email, backups. It read one application correctly and would have failed file utilities, database tools, point of sale, laboratory systems and most of the long tail Delphi is actually used for. A recogniser can only recognise what somebody already thought of.

Two things come out of reading an application:

- **Its own vocabulary**, with the furniture every program shares (File, Edit, Cancel, Save, OK, Close) set aside. What remains is characteristic: *invoice, ledger, posting, reconcile* for an accounts package; *rename, mask, extension, folder* for a file renamer.
- **Which sense of an ambiguous word this application means.** A shared list names the English words that are ambiguous in a user interface and gives each sense the evidence words that would be present if it were the one meant. The application's own vocabulary casts the vote.

*Volume* is loudness in an application that says mute and speaker; a disk in one that says partition and format. *Mask* is a filename pattern where the application talks about filenames. Nothing has to know what kind of application it is looking at.

Where the application settles nothing, **nothing is said**. A guess between two senses is worse than silence: it is a confident instruction to be wrong.

### 6.2 Part of speech

A button says what pressing it will do, so its caption is an instruction. A menu item names a thing. English hides the difference because its imperative and its dictionary form are the same word — so a service given *Help* alone may reasonably return a statement, and Arabic did: the third person singular, for *Help*, *Close* and *Play* alike. Grammatical, and useless on a button.

The context now states which is wanted, chosen from the control class, which every string already carries.



## 6.3 Getting the context to the string it describes

This was right and was being wasted. A service takes one context per request, and the client batched fifty strings and concatenated all fifty of their contexts into that single field — so *Help* arrived wearing forty-nine descriptions of other controls.

The economics make the fix free. **Billing is per translated character, not per request**, and DeepL does not count context characters at all. Measured on one application:

|                      | Characters |
|----------------------|------------|
| Source text (billed) | 6,471      |
| Context (not billed) | 65,466     |

The application costs 6,471 characters whether it goes as six requests or as 297. `DAT.Provider.Batching` therefore groups strings by *identical* context: where a context is unique the group holds one string; where many share a context, or have none, they travel together as before. The cost is round trips — a minute or two, once per language, for a result that is then stored.

## 7. The catalog and its validator

The catalog (`DAT.Core.CatalogJson`) is the editable record of one application in one language, held under `%LOCALAPPDATA%\DelphiAppTranslationStudio\Workspaces\<project>\Development`. Each entry carries the source text, the translation, where it came from, its classification, its context description and its status.

`DAT.Validation.Catalog` refuses to let a pack be built from a catalog with errors. It is not decoration. When the first Arabic run stopped with two blocking errors, those errors were real: the provider had turned every `%` into a `0`, so `%.2f GB used` came back as `0.2f`. Shipping that would have printed nonsense at users for ever. The validator is frequently the only thing standing between a plausible-looking translation and a broken application.

## 8. Translation providers

Two services are supported, both using the developer's own key: **DeepL** and **Google Cloud Translation**. The key is held in Windows Credential Manager or session-only, and never written to the catalog, the pack or any log.

Four units exist because four separate things went wrong in production. Each is worth knowing about before changing that code.

### 8.1 Placeholders (`DAT.Provider.Placeholders`)

Format specifiers are lifted out before a request and replaced by a token (`ZQPH0ZQPH`) that carries its own index, so a token the engine moves still returns as the specifier it stood for — right-to-left languages move things. Identical specifiers get separate tokens, because one string had three `%.2f` in it and a search-and-replace cannot tell them apart.

Afterwards the specifiers are restored and checked. If they still do not match, the **source text is returned** rather than the translation: an English string among Arabic ones is visible in review, whereas a broken format string is invisible until a customer sees it.

A string that is *nothing but* specifiers is never sent. `%.2d/%.2d` is a date, not a sentence.

### 8.2 Language codes (`DAT.Provider.LanguageCodes`)

A catalog names languages the way Windows does — `ar-SA`, `es-ES`. DeepL accepts two-letter codes plus a published list of variants (`EN-GB`, `EN-US`, `PT-BR`, `PT-PT`, `ZH-HANS`, `ZH-HANT`, `ES-419`, `DE-DE`, `DE-CH`, `FR-CA`, `FR-FR`) and rejects everything else with HTTP 400.

The earlier normaliser passed the region through unchanged, so German worked by luck — `DE-DE`

happens to be on that list — and Arabic, Spanish and Hebrew could never have worked. A region is now kept only where the service is known to accept it and dropped everywhere else. **Dropping is the safe direction:** the general code works for every language DeepL supports, so a language added after this was written still translates.

### 8.3 Retry (`DAT.Provider.Retry`)

A 429 is not a failure; it is the service asking for a slower pace, and the only wrong answer is to stop. Six attempts, waiting one second, two, four, eight, sixteen — thirty-one seconds of patience against the nine hundred milliseconds it used to allow. Retry-After is obeyed to the second where sent, capped, and treated as absent rather than guessed at when it is a date form or a zero.

A 400, 403 or 456 is *not* retried: a malformed request, a refused key and an exhausted quota all come back just as fast.

The client also paces itself. The delay starts at zero, grows by a quarter second each time a 429 is met and holds for the rest of the run — so a run that is never refused pays nothing, and one that is refused once slows down instead of arguing.

### 8.4 Saying what the service said

Refusals used to read "*Check the key, plan, billing, quota, language codes, and network connection*" — six possibilities, one of them right, and no way to tell which. Both services return an explanation in the body and it was being discarded. Rejections now quote it and add one line where the status code carries meaning of its own.

## 9. Terminology, dictionaries and hyphenation

Three shared, editable stores live under C:\Users\Public\Documents\Delphi App Translation. All three are plain JSON and are meant to be corrected by hand.

| Store                      | Scope        | Purpose                                                                  |
|----------------------------|--------------|--------------------------------------------------------------------------|
| Dictionaries\<lang>.json   | Per language | Approved wording earned on one application, available to every later one |
| Terms\ambiguous-terms.json | One, English | Words ambiguous in a UI, with the evidence that settles each sense       |
| Hyphenation\<lang>.json    | Per language | Where a long word may be broken                                          |

### 9.1 Hyphenation, and a measured difference between the frameworks

German builds one word where English uses three, and a single word cannot wrap: there is no space in *Benachrichtigungseinstellungen* for a column heading to break at, so it is simply cut off. Adding a language installs a companion dictionary saying which letters are vowels, which consonant groups are one sound, and how much of a word must be left whole at each end.

The pack carries a soft hyphen (U+00AD) at every allowed break, because nothing at build time knows how wide a control will end up. **Captions only** — format strings are left exactly as written, since their text goes on to be filled with data.

The two frameworks then differ, and this was measured rather than assumed:

- **FireMonkey honours soft hyphens.** A marked word measures exactly as wide as an unmarked one — 168.5 either way — and breaks at a mark when it must. Nothing needed doing.
- **GDI does not.** It draws U+00AD as an ordinary hyphen and DrawText will not break a line at one. The same word measured 205px marked against 170px plain, and stayed on one line.

So DAT.Runtime.VCL resolves the marks itself, after every layout rule has been applied and each control is the size it will really be: a control that wraps gets a real hyphen and a real line break at the last mark that fits; one that cannot wrap gets the plain word with the marks removed. **No soft hyphen ever reaches a VCL caption.**

# 10. The layout planner

DAT.Review.Localization is the largest and most consequential unit in the product. It decides what has to change so that translated text fits.

## 10.1 The measurement seam

Text is measured with the engine that will actually draw it: `GetTextExtentPoint32` through `DAT.Review.TextMeasurement.GDI` for the VCL, `TTextLayout` through `DAT.Review.TextMeasurement.FMX` for FireMonkey, chosen by `DAT.Review.TextMeasurement` from the catalog's framework. DPI is pinned to 96 because that is the basis a form was designed at.

Measuring with the wrong engine produces a plan that is confidently wrong — it fails correct layouts and passes incorrect ones — which is why this seam exists rather than one "good enough" measurement.

## 10.2 Framework facts the planner had to be taught

Each of these produced wrong layouts before it was understood:

- `TLabel.AutoSize` defaults to **True** and `WordWrap` to `False` — the opposite of an FMX label. A translated caption therefore stretches a VCL label before any rule is applied, so `AutoSize` must be cleared *before* the text is assigned, not after.
- A `.dfm` records fonts as `Font.Height` in negative pixels, not `Font.Size` in points. The conversion is `Round(-Height * 72 / 96)`, matching what `TFont` itself does, and fonts inherit down the object tree. Assuming twelve points where the form said nine overstated every caption by a third.
- The VCL names a second instance of a form `Name_1`. Looking up the instance name rather than the form's identity left every duplicate dialog untranslated.

## 10.3 The phases

The planner runs a sequence of passes over a single resolved model, so the values it finally exports agree with one another rather than describing conflicting placements:

1. **Phase 1** — start every control from its designer geometry.
2. **Phase 2** — size each control against its measured translated text; rows of buttons, captions above fields, and evenly pitched rows are treated as sets rather than individuals.
3. **Phase 3** — resolve collisions repeatedly against the planned geometry; hold captions inside the frames drawn around them.
4. **Phase 3d** — the settling pass: level rows, widen and re-centre headings, widen lone buttons rightward only (a button keeps its place), give stacked paragraphs one font size, narrow wrapped text to the width its wrap actually uses.
5. **Phase 3e** — mirror the form for a right-to-left language (section 11).

6. **Phase 4** — emit proposals from the settled geometry.

Every step in the settling pass is **trial and revert**: a change that breaks a contract is undone rather than kept.

## 10.4 Geometry the application owns

A late and important addition. `DAT.Review.CodeGeometry` reads the Pascal unit beside each form and notes any control whose `Left`, `Top`, `Width`, `Height`, `Position.X`, `Position.Y`, `Align` or `SetBounds` is assigned there. Those controls have their text translated and their **geometry left entirely alone**.

The case that produced it: an application centres its main heading in code at start-up, `lblMainHeader.Left := (Screen.Width - lblMainHeader.Width) div 2;`. The planner knew nothing of that, read the designer geometry, and proposed numbers that held the caption's own centre exactly — and overwrote a decision the application had already made and would never make again. Returning to English restored the *designed* position rather than the computed one.

The detection is deliberately literal: an assignment to a named identifier's geometry property and nothing cleverer. A control it misses behaves as before; a control it claims wrongly loses only its layout adjustments and is still translated. **Both directions fail softly**, which is the right property for a heuristic that reads somebody else's source.

One reading serves both frameworks, because what is read is Pascal rather than VCL or FireMonkey.

# 11. Right-to-left

Arabic, Hebrew, Farsi and Urdu were offered by the Wizard and marked `rt1` in the catalog long before anything read that value. A user could pick Arabic, get a clean run, and receive an application with Arabic text laid out left to right — worse than refusing, because it looks like it worked.

A right-to-left interface is a **reflected** interface, not reversed text.

## 11.1 Measured first

Two experiments settled the design before any code was written.

| BiDiMode                              | Flips alignment | RTL reading | Left scroll bar |
|---------------------------------------|-----------------|-------------|-----------------|
| <code>bdRightToLeft</code>            | <b>yes</b>      | yes         | yes             |
| <code>bdRightToLeftNoAlign</code>     | no              | yes         | yes             |
| <code>bdRightToLeftReadingOnly</code> | no              | yes         | no              |

A label set `taLeftJustify` draws as `DT_RIGHT` under `bdRightToLeft`. Since the planner already decides alignment for every control, that flip would land on top of ours and undo it silently.

**`bdRightToLeftNoAlign` is therefore what the runtime uses.**

The second experiment: digits look after themselves. A Hebrew word followed by 2026 comes out with the letters reversed and the year intact, in both renderers. Version strings, times and quantities need no help.

## 11.2 Mirrored by the planner, not by the framework

The VCL has `BiDiMode` and `FlipChildren`. FireMonkey has neither. Leaning on the VCL's mechanism would mean writing the FireMonkey half separately and getting different behaviour on each, so the mirror is computed by the planner and emitted as the ordinary position rules the runtime already applies. The transform is parent-relative, which handles nesting without recursion:

`MirroredLeft := ParentInnerWidth - (PlannedLeft + PlannedWidth)`

**A form is never mirrored.** A window has no parent to be reflected within, so the arithmetic degenerates into negating its own position — which put three forms of one application at a negative `Left` and showed a strip of desktop down one side.

## 11.3 What mirrors, and what deliberately does not

| Mirrors                                                                                                                                                                                                                                       | Does not                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coordinates, within each parent<br>Align, for framework-placed controls<br>Anchors, where one horizontal edge is anchored<br>Text alignment (centre stays centre)<br>Grid column order<br>Tab order<br>Reading order and scrollbar side (VCL) | Transport buttons — rewind, play and stop refer to tape direction, not reading direction, so the group moves as a block and keeps its order<br><br>Numbers, times, versions and paths — handled by the renderers<br><br>Images — the transform only ever moves a control, never its contents, so logos are safe by construction |

## 11.4 Order of application

Reading order is applied **before** the text, and the reason is subtle enough to document. Translating a menu item's caption rebuilds the menu, and Delphi stamps each item with the reading order in force at the moment of the rebuild. Applying direction after the text meant rebuilding the menu in the language being *left*, then relying on `TMenu.DoBiDiModeChanged` to correct it — which returns early on several conditions, including a window handle that is momentarily zero while the form's window is recreated. Menus stayed reversed until the application was restarted.

Setting direction first removes the dependency entirely: whatever is rebuilt afterwards is rebuilt the right way round to begin with.

Worth knowing when reproducing menu behaviour: `TMenu.DoBiDiModeChanged` begins if (not `SysLocale.MiddleEast`) or (`WindowHandle = 0`) then `Exit`. On a machine whose Windows locale is not Middle-Eastern, the VCL does not lay menus out right-to-left at all, whatever `BiDiMode` says.



## 12. The runtime pack

`DAT.Core.RuntimePack` writes schema 3: language and locale, the translated strings by key, source-text fallbacks, runtime templates, font colours, and the layout rules. It is the only artefact that ships.

Two rules govern what may go in:

- A proposal must be **accepted**. A rejection made in review is honoured; a proposal left pending is not exported.
- A property must be in the one list — `IsRuntimeLayoutProperty` in `DAT.Runtime.LanguagePack`.

Both rules once had private copies elsewhere and both silently deleted features. The pending case is the more instructive: `RestoreDecisions` reads the previous run's proposal file and copies each saved decision over the analyser's, so that a rejection survives a re-scan. But it also copied *pending*, and pending is not a decision — it is the absence of one. A proposal file written by a build that did not know a property existed recorded it as pending, and every later run restored that over a freshly accepted decision. The feature was vetoed in perpetuity by a stale file, and no amount of rebuilding the analyser could have fixed it.

## 13. The runtime

`DAT.Runtime.VCL` and `DAT.Runtime.FMX` apply a pack to a live form. `DAT.Components.Core` holds the shared language manager; the VCL and FMX adapters differ only where the frameworks do.

### 13.1 Order of operations in `ApplyToForm`

1. Snapshot the original geometry — once, before any language is applied.
2. **Restore** from that snapshot. Without this, each language inherits whatever the last one changed, and a rule simply absent from the new pack has nothing to undo it.
3. Apply reading order (section 11.4).
4. Apply text, including collections such as grid headings.
5. Apply layout rules in a fixed order: `AutoSize`, `FontSize`, `WordWrap`, `alignment`, `Align`, `size`, `position`, `Anchors`, `TabOrder`, column widths, column order.
6. Resolve soft hyphens, now that every control is its final size.

The ordering is not arbitrary. An auto-sizing label recomputes its own bounds from one line of text and discards an assigned width, so `AutoSize` must be cleared first. Column widths name a column by its designed index, so the column *order* must be reversed last or every width lands on the wrong column.

## 13.2 What is never restored

**Colour.** The applicator does not set colours, so putting one back can only undo what the application itself did. One application paints its own colours from a menu after the form is up; restoring the design-time colour stamped a maroon heading over a white one and turned captions black. What we never changed, we never restore.

## 14. Contracts — how this product is kept honest

This section describes the practice that most distinguishes this codebase. It is worth reading even by someone who intends to change nothing.

### 14.1 What a contract is

A layout contract is three files: a small purpose-built form, a catalog naming the translated text for it, and an expectation file stating in numbers what the analyser must do. Assertions are numeric because layout is numeric — a right edge that must not move, a control that must keep its place, a caption that must still hold its text.

The forms are purpose-built rather than taken from a real application, so each rule is proved on the *shape* of a problem rather than on one customer's project.

**54 layout contracts** currently run, alongside 3 form-scan fixtures, 9 pascal-scan checks and **28 test harnesses**.

### 14.2 A contract must fail first

The single most important rule. A contract whose expected values are copied from what the code currently does proves nothing — it records behaviour instead of requiring it.

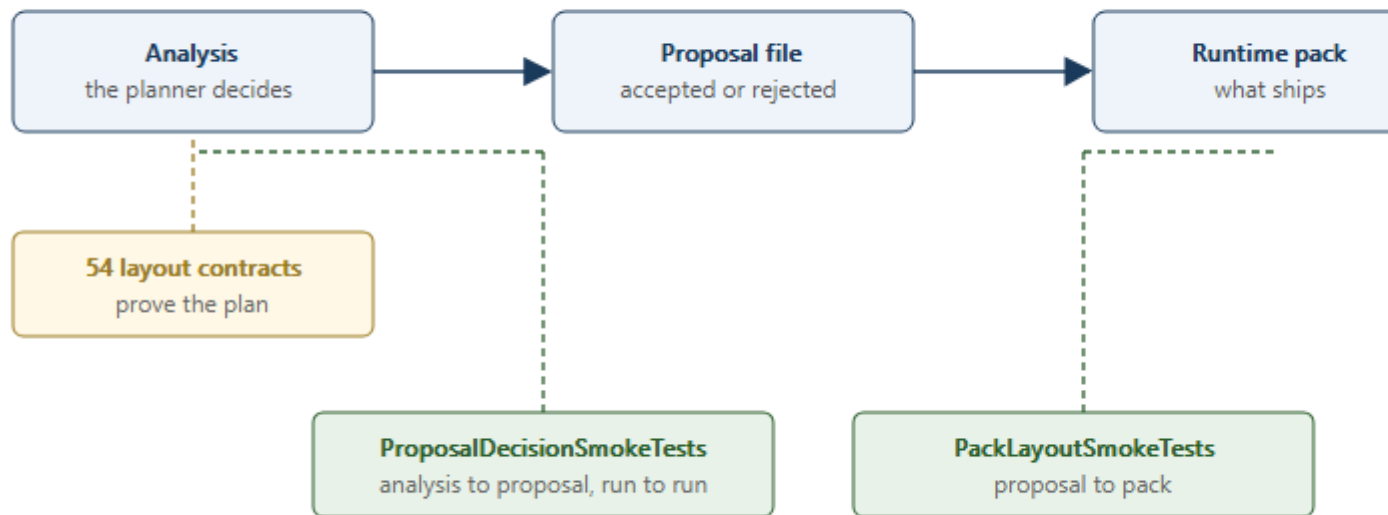
Two contracts in this project had to be rebuilt for exactly that fault: one flipped a constant the fixture never reached, and another used a caption that lacked the hard line breaks the real one had. Both passed. Both were worthless.

The discipline is therefore: write the assertion, watch it fail for the right reason, then make it pass. Where a test was written after the code — which happens — it is verified by deliberately disabling the code and confirming the test fails. This guide's author did that twice in one day.

## 14.3 Contracts must cross seams, not sit at ends

**Figure 2 — Where the tests sit, and the seams between them**

A value can be correct at both ends of the pipeline and absent in the middle. Three defects hid in exactly that gap.



Tests at the ends pass while the middle is empty. Only a test that crosses a seam can see the hole.

The hardest defects this product has had all shared one shape: **a value correct at both ends of the pipeline and absent in the middle.**

Three separate failures deleted the right-to-left feature before anyone noticed:

1. The FireMonkey applicator held its own copy of the allowed-property list and dropped every mirroring rule.
2. The pack exporter held another copy and dropped them again.
3. The auto-accept rule held a third, so the proposals were created *pending* — and `RestoreDecisions` made that permanent.

Every one was invisible to every test that existed, because the tests sat at the ends: the layout contracts passed because the plan was right, and the applicator tests passed because they were handed a pack written by hand. Nothing crossed the joins.

Three tests now do: `PackLayoutSmokeTests` goes proposal-to-pack, the contract harness can assert a proposal's *decision* as well as its value, and `ProposalDecisionSmokeTests` goes run-to-run. The first failed on seven of eight checks when written.

## 14.4 A test must not change what it measures

A field diagnostic added to the VCL runtime read `TMenu.Handle` to report the native right-to-left flag. `TMenu.Handle` creates and populates the menu if it does not exist — and doing that mid-apply **destroyed the translated menu captions**. The VCL runtime smoke test failed within seconds of the

diagnostic being switched on, which is exactly what that test is for. The diagnostic was changed to ask Windows for the menu already on the window, which creates nothing.

## 14.5 Guards around the suite

- **Stale-harness rejection.** If any analyser source is newer than the contract executable, the run stops rather than reporting on code that is no longer the code being changed.
- **Coverage enforcement.** A fixture with no expectation file fails the run, so a rule cannot quietly exist without a contract.
- **Source encoding.** `tools\check_source_encoding.ps1` fails the build if any `.pas`, `.dpr` or `.dpk` holds a character outside ASCII without a byte order mark — because Delphi reads such a file in the ANSI codepage and its text arrives wrong. It has caught two real defects that would otherwise have shipped, including the German and Spanish vowel sets in the hyphenation dictionaries.
- **Rebuild after every change.** All four configurations, every time. An evening was once lost to testing a three-hour-old binary.

## 14.6 Framework parity — an honest gap

The 54 contracts cover **33 distinct behaviours**: 21 proven on both frameworks, **12 on VCL only, none on FireMonkey only**. Ten of the twelve are framework-neutral planner behaviour that simply has no FireMonkey fixture yet; one (`inherited_font_is_the_forms_font`) is genuinely VCL-specific by design, and one is partly so.

Because the planner measures through a seam, the same rule can legitimately produce different numbers on each framework — which is the argument for a twin rather than an assumption. Appendix B lists them. They must be written one at a time, each failing first; producing them in bulk from current output would recreate precisely the fault described in 14.2.

## 15. Build and release validation

All four configurations — Win32 and Win64, Debug and Release — are rebuilt after every change. `tools\tests\RunPhase10ReleaseValidation.ps1` requires one uninterrupted pass of the complete matrix, plus the Studio launch and self-localization smoke tests.

The full suite: 54 layout contracts, 3 form-scan fixtures, 9 pascal-scan checks, the foundation, scanner, catalog, runtime, validation and export tests, the VCL and FMX runtime smoke tests, the four language-manager suites, and the focused harnesses for retry, language codes, context batching, proposal decisions, pack export, placeholders, hyphenation, context, wrap, discovery, and right-to-left on both frameworks.

Runtime **packages** must be rebuilt separately when a runtime unit changes; an application that links them will not otherwise pick the change up. This has cost real debugging time and is easy to forget.

## 16. Known limits

Recorded properly in `docs\guides\Fix List.md`. The ones an engineer should know before planning work:

- **No translation memory** and no TMX/TBX interchange. This is the largest functional gap against every commercial competitor and the main obstacle to a customer migrating existing assets in.
- **Runtime-composed strings** cannot be translated without a call in the target application, and the product does not yet tell the developer which strings those are — though it holds the file and line for each.
- **Rows at a fixed pitch cannot grow.** Where a language needs more width per member than the designed pitch allows, the last members are wrapped or cramped rather than the row being re-pitched.
- **First run shows the source language** whatever the machine is set to; nothing consults the operating system's UI language.
- **CJK is untested.** Chinese, Japanese and Korean do not wrap on spaces and do not hyphenate, and they usually run shorter than English rather than longer — so the settling pass would be exercised in a direction it has never seen.
- **Validation breadth.** Every significant defect this product has found came from a real application. At the time of writing that is one application plus a synthetic test app, and the observed rate suggests a second and third will find another batch.

## Appendix A — Unit inventory

| Unit                                                        | Purpose                                                |
|-------------------------------------------------------------|--------------------------------------------------------|
| <b>core</b>                                                 |                                                        |
| DAT.Core.Types                                              | Catalog, entry, locale and enumeration types           |
| DAT.Core.CatalogJson                                        | Catalog read and write                                 |
| DAT.Core.RuntimePack                                        | Builds the shipped pack                                |
| DAT.Core.Glossary                                           | Per-project approved terms                             |
| DAT.Core.SharedDictionary                                   | Per-language wording shared across applications        |
| DAT.Core.Hyphenation                                        | Per-language break dictionaries                        |
| DAT.Core.Terminology                                        | Authoritative term resolution                          |
| DAT.Core.TranslationWorkspace                               | Where every artefact lives                             |
| DAT.Core.ProjectDetection                                   | Recognising a Delphi project and its framework         |
| DAT.Core.AITranslation                                      | The copy-and-paste AI workflow                         |
| <b>scan</b>                                                 |                                                        |
| DAT.Scan.Project                                            | Drives the scan                                        |
| DAT.Scan.FormText                                           | Reads .dfm and .fmx                                    |
| DAT.Scan.PascalResources                                    | Reads .pas                                             |
| DAT.Scan.Context                                            | Writes each string's context sentence                  |
| DAT.Scan.DomainProfile                                      | Vocabulary and word-sense resolution                   |
| DAT.Scan.Rules / Quality / TextCodec / Types / CatalogMerge | Classification, quality checks, encoding, types, merge |
| <b>review</b>                                               |                                                        |
| DAT.Review.Localization                                     | The layout planner                                     |

|                                                        |                                             |
|--------------------------------------------------------|---------------------------------------------|
| DAT.Review.TextMeasurement                             | The measurement seam                        |
| DAT.Review.TextMeasurement.GDI / .FMX                  | Per-framework measurement                   |
| DAT.Review.CodeGeometry                                | Controls the application positions itself   |
| <b>provider</b>                                        |                                             |
| DAT.Provider.Client                                    | DeepL and Google                            |
| DAT.Provider.Placeholders                              | Format specifier protection                 |
| DAT.Provider.Batching                                  | Grouping by shared context                  |
| DAT.Provider.LanguageCodes                             | Codes a service will accept                 |
| DAT.Provider.Retry                                     | Rate limits and backoff                     |
| DAT.Provider.CredentialStore / Settings / Types        | Keys, settings, provider types              |
| <b>runtime (ships in the customer's application)</b>   |                                             |
| DAT.Runtime.LanguagePack                               | Pack loading; the one allowed-property list |
| DAT.Runtime.VCL / .FMX                                 | Applying a pack to a live form              |
| DAT.Runtime.Manager                                    | Language selection and format settings      |
| DAT.Runtime.Preference                                 | Remembering the chosen language             |
| <b>components (ship in the customer's application)</b> |                                             |
| DAT.Components.Core                                    | Shared language-manager behaviour           |
| DAT.Components.VCL / .FMX                              | Framework adapters                          |
| DAT.Components.VCL/FMX.LanguageSelector                | Optional bound selector                     |

## Appendix B — Contract inventory and framework parity

21 behaviours proven on both frameworks: button above grid padding; button keeps its place when text grows; button row keeps even pitch; caption above field takes its column; centred heading widens about



centre; checkbox caption inside container; checkbox caption note; code-positioned control is left alone; container keeps its size; designed overlap is left alone; email label button padding; frame grows and says so; grid headers fit; intro paragraph wraps compact; left caption takes left margin; long button and field stack; media button row container; memo label pair; preserve label font; right-aligned caption grows leftward; right-to-left mirrors the form.

**Twelve awaiting a FireMonkey twin:** a button gets room for its caption; a caption stops at the button beside it; a grid does not end the form; a heading widens before it wraps; a long word gets room; a paragraph stays on the form; caption far wider than its box; stacked paragraphs share a size; transport buttons keep their order; wrapped text is balanced; right-to-left flips alignment and anchors (*partly VCL-specific — Anchors is a VCL notion*); inherited font is the form's font (*genuinely VCL-specific — font inheritance was changed for the VCL only, and FireMonkey's differing defaults were deliberately left alone*).

## Appendix C — Files, folders and where things live

| What                                    | Where                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------|
| Development catalog                     | %LOCALAPPDATA%\DelphiAppTranslationStudio\Workspaces\<project>\Development                   |
| Runtime packs                           | ...\Workspaces\<project>\Languages, deployed to Localization\Languages beside the executable |
| Layout proposal and review              | export\localization-review\<project>\<language>                                              |
| Shared dictionaries, terms, hyphenation | C:\Users\Public\Documents\Delphi App Translation                                             |
| Language preference (target app)        | %LOCALAPPDATA%                                                                               |
| Contracts                               | contracts\layout                                                                             |
| Test harnesses                          | tools\tests                                                                                  |
| Engineering notes and Fix List          | docs\guides                                                                                  |

*This guide describes the product as it stood on 20 August 2026. Where it disagrees with the code, the code is right and this document is stale — the Engineering Notes in docs\guides\Engineering Notes.md carry the running record of change.*